Long arm metal microdrive

Misi Voroslakos

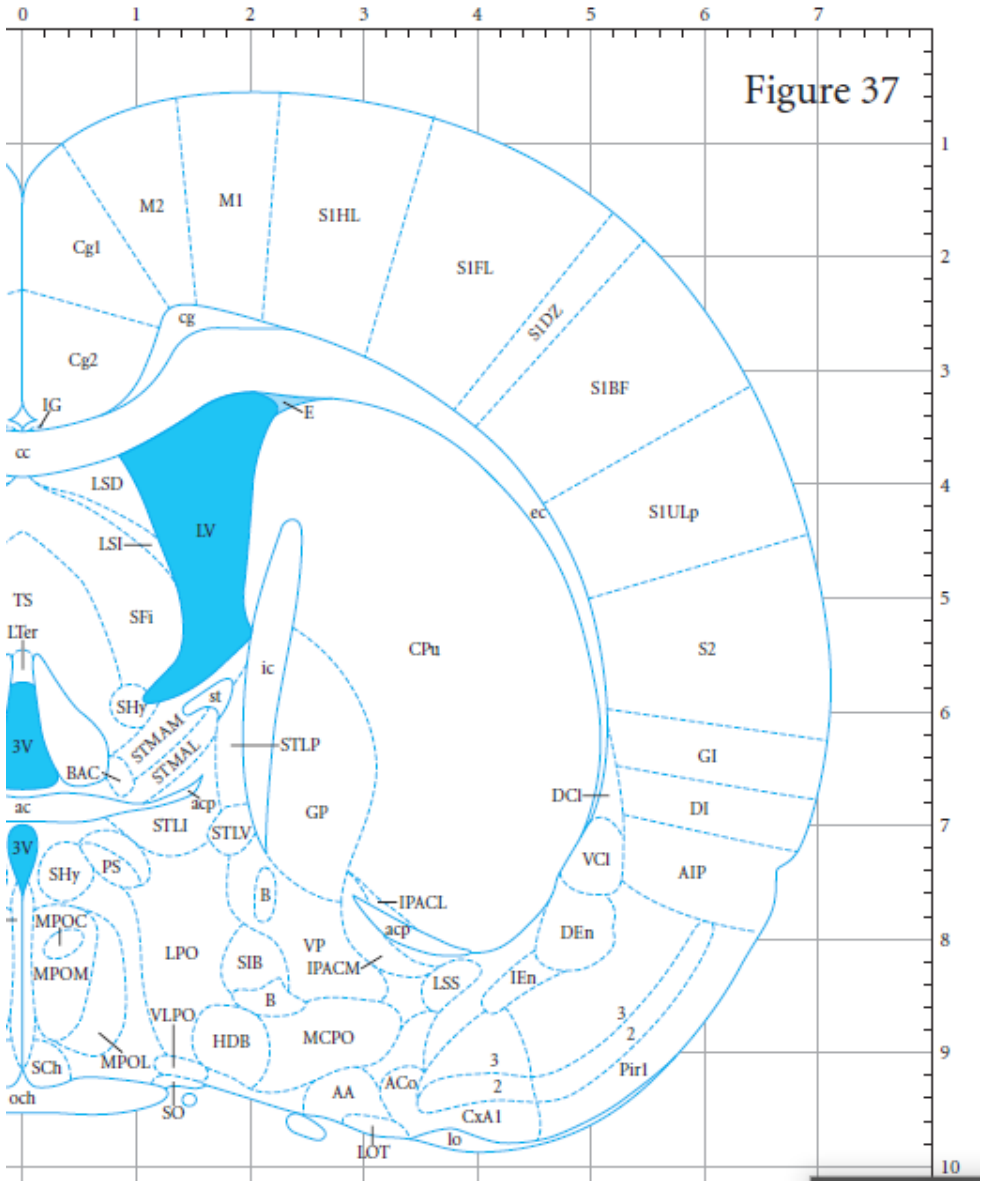
2026. 10. 19.

Issue with current design

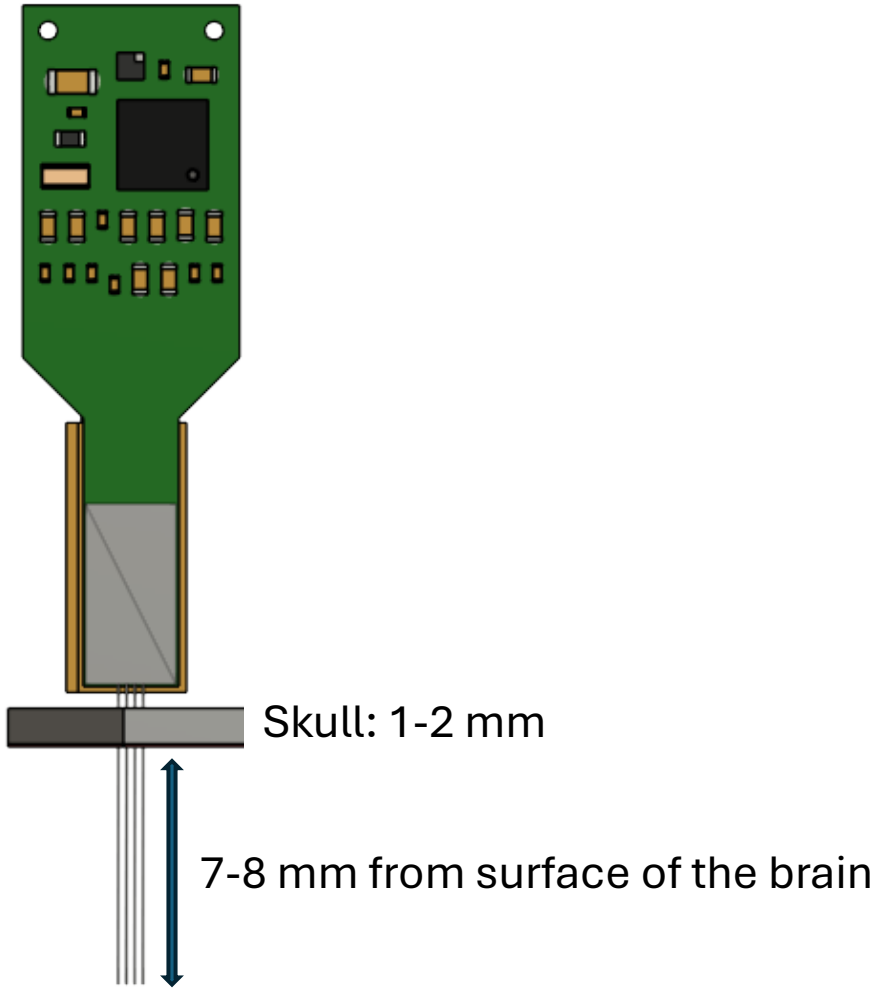
- NPX probe's shank is 10 mm
- Rat brain is ~9-10 mm in DV

- **We need space for the skull**
  - **1-2 mm**

- Deep targets cannot be reached.



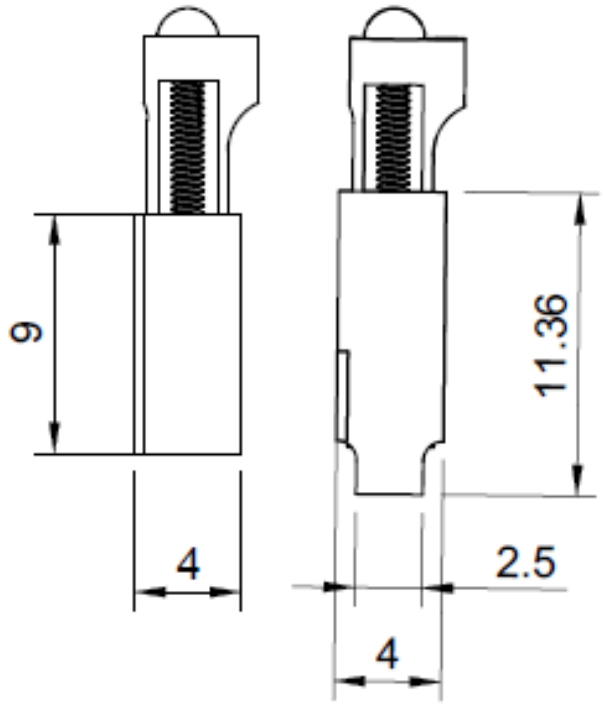
Skull issue explained

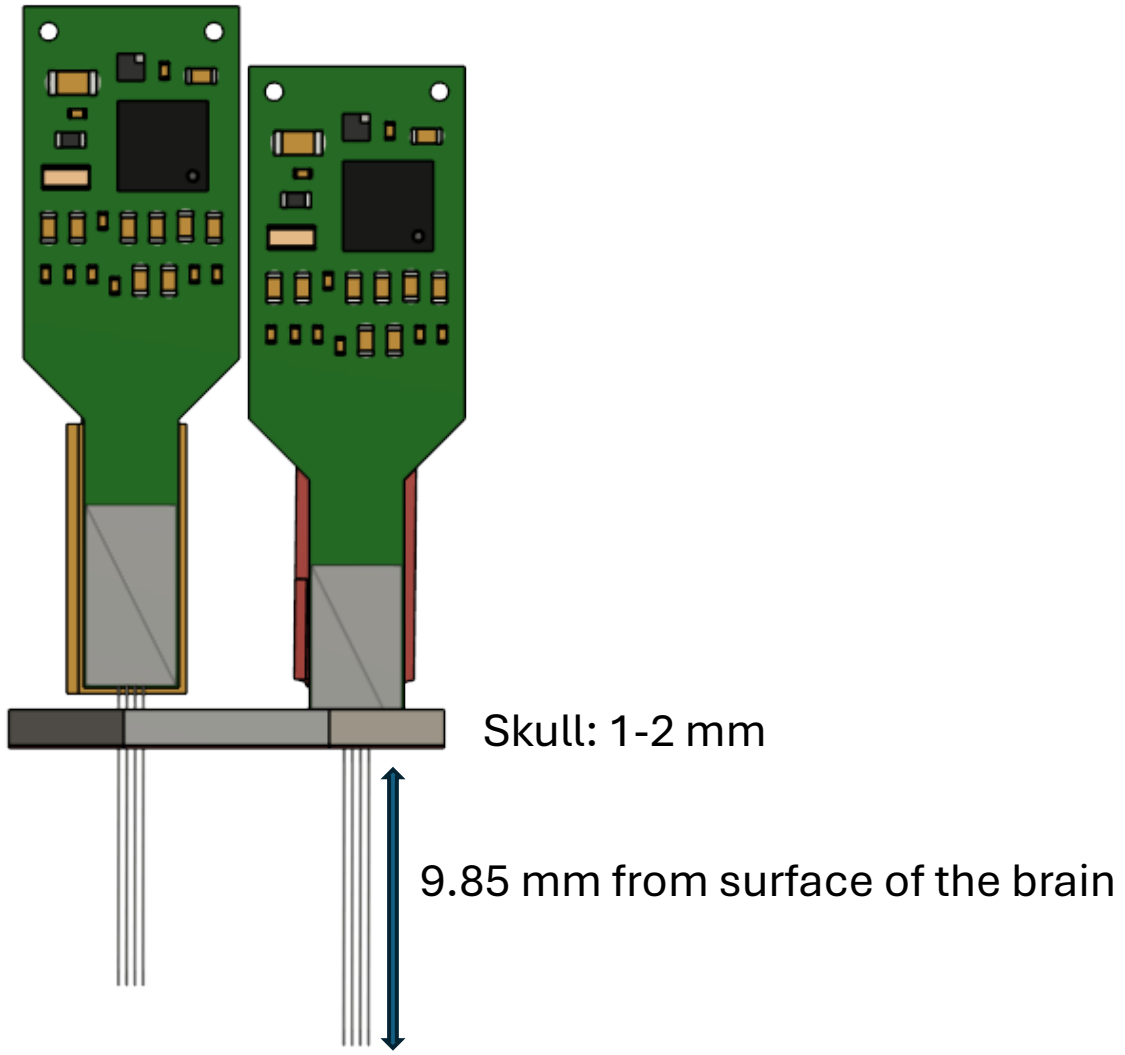


- This issue is more pronounced if probe is implanted with an angle.
- We also lose top recording channels capability as it can never reach the brain.

Solution

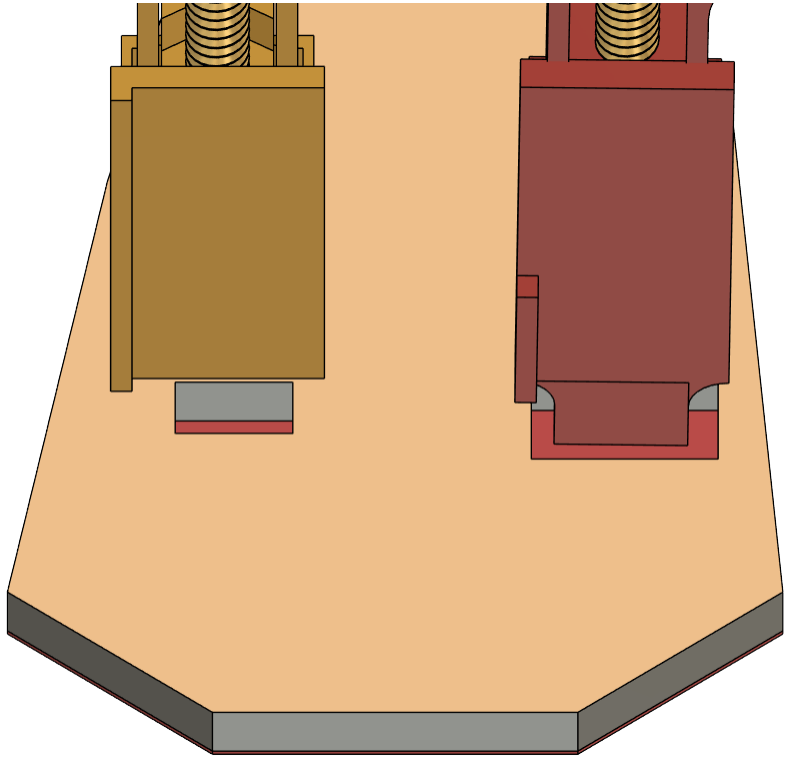
- Tapered Arm design.

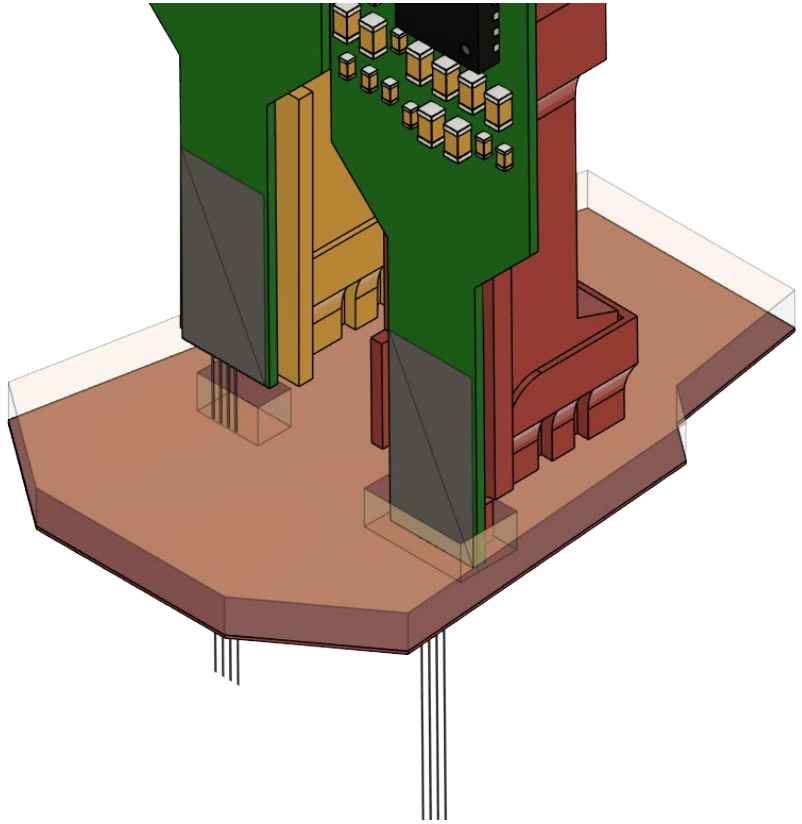


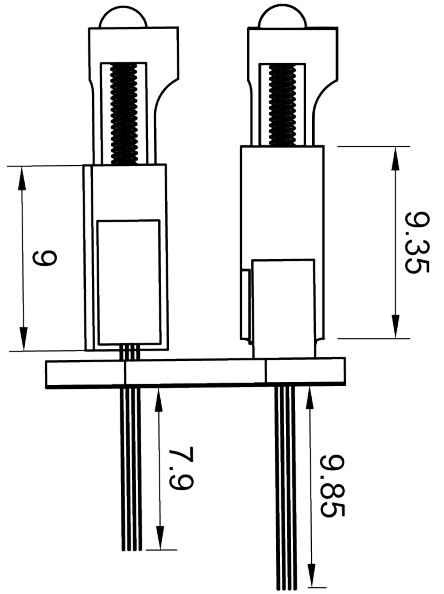


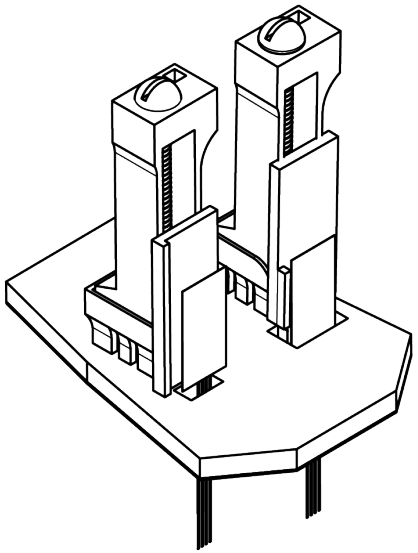
- Create a bigger craniotomy.
- Arm can “enter into” the skull.

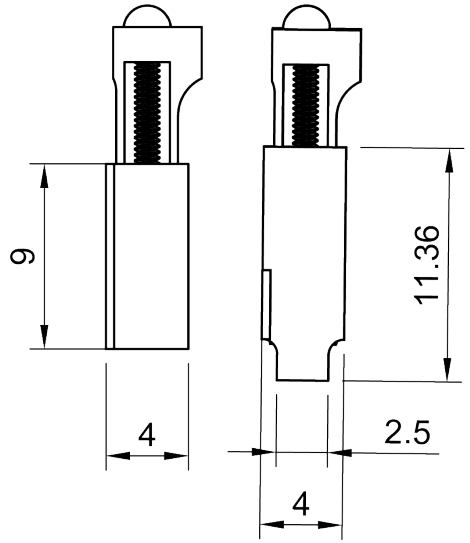
- Left – usual microdrive.
- Right – tapered arm inside skull.











Dept.	Technical reference	Created by Mihaly Voroslakos 10/19/2026	Approved by	
		Document type	Document status	
		Title long_arm_explained	DWG No.	
			Rev. / Date of issue	Sheet 1/1